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A student determined the dry mass of lettuce leaves. The equipment used was an oven and a weighing balance.

Describe how the student could use this equipment to determine the dry mass of lettuce leaves.

[2 marks]

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The student then used a calorimeter to determine the chemical energy stored in a sample of lettuce leaves with a dry mass of 1.50 g

The 1.50 g sample was fully combusted in a calorimeter.

The volume of water in the calorimeter was 150 cm³

The increase in temperature of the water was 18.2 °C

4.18 J of heat energy are required to increase the temperature of 1 cm³ of water by 1 °C

Use this information to calculate the heat energy released, in kJ g⁻¹, of dry mass of lettuce leaves.

Show your working.

[2 marks]

Answer _____ kJ g⁻¹

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Biomass can be measured as the dry mass of tissue per given area. Some crop plants are grown in large greenhouses (glasshouses) where optimum growing conditions can be maintained.

Growing crop plants at higher temperatures than their optimum temperature can decrease their biomass.

Use your knowledge of net primary production to explain why.

[2 marks]

Turn over for the next question

